UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

PART A (10 MARKS)

Answer ALL questions.

1. Which statement immediately exits a loop?
   A. cin
   B. return 0
   C. continue
   D. break

2. Which declaration creates a floating-point variable named price?
   A. char price;
   B. int price;
   C. bool price;
   D. float price;

3. What does a compiler do?
   A. Executes one source statement at a time
   B. Translates the whole source program before execution
   C. Draws flowcharts automatically
   D. Stores only integer values

4. Which identifier is valid in C++?
   A. total score
   B. total_score
   C. 2total
   D. int

5. Which declaration creates an integer array of five elements?
   A. int a(5);
   B. int a;
   C. int a[5];
   D. array a[5];

6. Which statement is most suitable for selecting among several constant cases?
   A. switch
   B. do...while
   C. for
   D. while

7. Which part of a for loop is executed after each iteration?
   A. The loop body twice
   B. The update expression
   C. The initialisation
   D. The condition only

8. Which phase of the program development life cycle defines what the program must do?
   A. Maintenance
   B. Coding
   C. Analysis
   D. Testing

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9. Which keyword prevents a variable's value from being changed after initialisation?
   A. const
   B. static
   C. void
   D. break

10. Which operator sends data to cout?
   A. >>
   B. =
   C. ==
   D. <<

PART B (60 MARKS)

QUESTION 1

This question tests algorithms and programming errors.

a) Define algorithm and pseudocode.

(2 marks)

b) Explain THREE types of programming errors and give one example of each.

(3 marks)

QUESTION 2

A program reads three integer values and displays the largest value.

a) Draw a flowchart to find the largest of three values.

(8 marks)

b) Write pseudocode and a complete C++ program for the flowchart.

(5 marks)

QUESTION 3

This question tests type conversion and constants.

a) Explain the difference between implicit and explicit type conversion, and give one C++ example of each.

(6 marks)

b) Write a C++ program segment that calculates the average of three integer marks as a double and displays it to two decimal places.

(5 marks)

QUESTION 4

Study the following nested loop:

```
for (int i = 1; i <= 3; i++) {
   for (int j = 1; j <= i; j++) {
      cout << j;
   }
   cout << endl;
}
```

a) Trace the program and show the exact output.

(6 marks)

b) Identify TWO logic errors in a version that prints four rows but reuses j without resetting it. Rewrite the corrected code.

(9 marks)

QUESTION 5

A shop stores the prices of five items in an array.

a) Declare an array named prices with the values 20.5, 55.0, 15.9, 75.0 and 10.0. Display all values.

(6 marks)

b) Write a program segment that applies a 10% discount to prices above RM50 and stores the results in a second array. Display both arrays.

(10 marks)

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PART C (30 MARKS)

QUESTION 1

A bookstore records up to five books. Each book has a title, price and quantity in stock.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

QUESTION 2

A cafe sells three items: coffee RM8.50, tea RM6.00 and juice RM9.50. A 10% discount is given when the subtotal is at least RM50.

Write a complete C++ program that reads the item code and quantity, calculates the subtotal and discount, and displays a formatted receipt with the item name, quantity, subtotal, discount and amount to pay.

(15 marks)

END OF QUESTION PAPER